

Objects 3

Brainster Web Development Academy



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Classes

- Since 2015 (ES6), JavaScript also has classes. But aside from the syntax, they are in no way different from object prototypes.

```
function Cat(nameArg, colorArg) {  
  this.name = nameArg;  
  this.color = colorArg;  
}
```

```
class Cat {  
  constructor(nameArg, colorArg){  
    this.name = nameArg;  
    this.color = colorArg;  
  }  
}
```

```
const cat1 = new Cat('Thomas', 'blue');
```

```
const cat1 = new Cat('Thomas', 'blue');
```



Classes in JavaScript

- It doesn't look too different, it doesn't behave different but it can be more readable when it comes to having methods in the class.
- Otherwise, it's a matter of choice - there is no "right" choice, as long as you understand how it works.
- Most popular libraries / frameworks support both ways of doing it, although classes tend to be a bit more used.

```
3 class Cat {  
    constructor(nameArg, colorArg){  
        this.name = nameArg;  
        this.color = colorArg;  
    }  
  
    sayNameAndColor() { // notice the lack of 'this' and the lack of the function keyword  
        alert(this.name + ' ' + this.color);  
    }  
}  
  
const cat1 = new Cat('Thomas', 'blue');
```



Class methods

- When it comes to methods, unlike in object prototypes:
 - We don't need to write "function" before the declaration
 - We don't need to write "this." before declaring the function - it's implied when it comes to a class.
 - If you want "this" to work inside the method, DO NOT USE an arrow function (we'll get to "why" next lesson).



Exercise 1

We had an exercise about cats some time ago. Let's rewrite it using object classes.

1. Create a new **class** Cat that takes two inputs: name and color.
2. Add a new method to it called "sayNameAndColor" which alerts the name and color of the cat.
3. Add another method called "setName" which changes the name of the cat.
4. Create a new cat instance (e.g. "Thomas", "blue").
5. Invoke sayNameAndColor.
6. Change the cat's name via setName.
7. Invoke sayNameAndColor again.



Classes and other types

- Classes by themselves aren't too useful (just as object prototypes).
- However, the objects the produce when we call the new keyword (**instances**), are just regular objects.
- That means, we can interact them in the same way as with any other object. We can change the properties of those objects, put them in arrays etc.



Exercise 2

We had an exercise about dogs some time ago. Let's rewrite it using object classes.

1. Create an empty array called "dogs".
2. Create a **class** Dog which takes name and color arguments.
3. (In html) create two input fields (with different ids) and a button that says "Add dog".
4. Create a function which:
 - a. Gets the values from the two input fields
 - b. Creates a new "dog" **instance** using the first value as name and the second as color.
 - c. Adds the newly created dog object to the array.
 - d. Logs in the console the whole "dogs" array
5. Invoke the function when the button gets clicked (using "click" with addEventListener)



Exercise 3

1. Continue on the previous exercise: whenever a new dog is entered, clear the value from the inputs.



Exercise 4

1. Continue on the previous exercise: Add an extra column to the table which contains an “edit” button.
2. Define an event listener for the edit button - when clicked, it allows us to edit the dog’s name and color. It does it by filling the inputs with the existing values for that dog, but also remembers which record we’re trying to edit.
 - Hint: Keep some variable somewhere which tells the code whether a dog is being entered as new, or an existing one is being edited.
 - Hint: Don’t rush it - coding is 80% thinking and 20% writing actual code ;)



Homework

Expand the previous exercise further:

1. Add a select box which lets you select a cat or a dog.
2. Create a cat class, similar to the dog class, with the same methods.
3. When we click the “add” button, we should either add a new dog or a new cat based on the selection from the select box.
4. The edit functionality should continue working regardless of whether it's a cat or a dog that we're editing.

Hint: It's not easy. Plan a solution before starting to code :)

